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#include <stdio.h>

#include <mpi.h>

int main(int argc, char** argv)
{
    int rank, size;

    int send_data[4] = {10, 20, 30, 40};

    int recv_data;

    MPI_Init(&argc, &argv);

    MPI_Comm_rank(MPI_COMM_WORLD, &rank);

    MPI_Comm_size(MPI_COMM_WORLD, &size);

    // Scatter 1 integer from send_data array on Root (rank 0) to each process's recv_data
    MPI_Scatter(send_data, 1, MPI_INT, &recv_data, 1, MPI_INT, 0, MPI_COMM_WORLD);

    printf("Process %d received: %d\n", rank, recv_data);

    // Each process modifies its received chunk
    recv_data += 1;

    // Gather 1 modified integer from each process back into send_data array on Root (rank 0)
    MPI_Gather(&recv_data, 1, MPI_INT, send_data, 1, MPI_INT, 0, MPI_COMM_WORLD);

    // Root process prints the newly gathered data
    if (rank == 0)
    {
        printf("\nGathered data on Process 0: ");

        for (int i = 0; i < size; i++)
        {
            printf("%d ", send_data[i]);
        }
    }
}

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    printf("\n");  
}  
  
MPI_Finalize();  
return 0;  
}
```